Zigbee2MQTT BLZ Installation Guide

BLZ support for Zigbee2MQTT follows the same setup as standard Zigbee2MQTT — the only differences are:

1. **Use the BLZ repositories** listed below (instead of the official ones)
2. **Set `adapter: blz` and `baudrate: 2000000`** in your serial configuration

This guide covers the three most common installation methods. For other platforms (openHABian, Windows, FreeBSD, Kubernetes, etc.), follow the official Zigbee2MQTT installation guide and substitute the repositories and serial settings from this document.

BLZ Repositories

Component	BLZ Repository
Z2M Home Assistant App	pine64/hassio-zigbee2mqtt
Z2M Docker image	`ghcr.io/pine64/zigbee2mqtt:latest-dev`
Z2M source (BLZ branch)	pine64/zigbee2mqtt `feat/blz-local-dev`
zigbee-herdsman (BLZ branch)	pine64/zigbee-herdsman `feat/blz`
Dongle firmware & flashing	ThirdReality Zigbee 3.0 USB Dongle

BLZ Serial Configuration

All installation methods use the same serial configuration. The only difference from a standard Zigbee2MQTT setup is the `serial` section:

```yaml
serial:
  port: /dev/ttyUSB0   # your dongle path
  adapter: blz
  baudrate: 2000000
```

A complete `configuration.yaml` example:

```yaml
mqtt:
  base_topic: zigbee2mqtt
  server: mqtt://localhost:1883
serial:
  port: /dev/ttyUSB0
  adapter: blz
  baudrate: 2000000
frontend:
  enabled: true
  port: 8099
homeassistant:
  enabled: true         # set false if not using HA
```

For all other configuration options, see the Zigbee2MQTT configuration reference.

Option 1: Home Assistant App (HAOS)

Prerequisites

- Home Assistant OS (2026.2+) with Supervisor

- Mosquitto broker app installed (from the HA App Store)

Steps

1. Go to **Settings** > **Apps** > **App Store**
2. Click the three-dot menu (top right) > **Repositories** and add:

   `https://github.com/pine64/hassio-zigbee2mqtt`
3. Refresh, find **Zigbee2MQTT Edge - BLZ** and click **Install**
4. In the app's **Configuration** tab, set:
   - Serial port: `/dev/ttyUSB0` (or your dongle path)
   - Adapter: `blz`
   - Baud rate: `2000000`
5. Start the app
6. Open the Zigbee2MQTT frontend from the sidebar (or at `http://<HA_IP>:8099`)

For general troubleshooting, see the official Z2M Home Assistant guide.

Option 2: Docker

Prerequisites

- Docker installed
- An MQTT broker (e.g., Mosquitto) running and accessible

Steps

1. Create a data directory and `configuration.yaml` (see BLZ Serial Configuration above)
2. Run the container:

```
docker run -d \
  --name zigbee2mqtt-blz \
  --restart=unless-stopped \
  --device=/dev/ttyUSB0:/dev/ttyUSB0 \
  -p 8099:8099 \
  -v $(pwd)/z2m-data:/app/data \
  -e TZ=UTC \
  ghcr.io/pine64/zigbee2mqtt:latest-dev
```

3. Open the frontend at `http://localhost:8099`

For general Docker setup, see the official Z2M Docker guide.

Option 3: From Source

Prerequisites

- An MQTT broker (e.g., Mosquitto) installed and running
- Git and Node.js (v20+) with pnpm installed

For a full overview, see the official Z2M Linux installation guide.

Step 1: Clone and build zigbee-herdsman (BLZ fork)

```
git clone -b feat/blz https://github.com/pine64/zigbee-herdsman.git
cd zigbee-herdsman
pnpm install
cd ..
```

Step 2: Clone and build Zigbee2MQTT (BLZ fork)

```
git clone -b feat/blz-local-dev https://github.com/pine64/zigbee2mqtt.git
cd zigbee2mqtt
pnpm install
pnpm run build
```

Step 3: Configure and start

Create `data/configuration.yaml` (see BLZ Serial Configuration above), then:

```
pnpm start
```

Additional Notes

- BLZ adapter uses **2M baud** (2,000,000) serial communication
- Ensure your MQTT broker is running before starting Zigbee2MQTT
- The BLZ dongle firmware must match the protocol version supported by zigbee-herdsman
- For firmware flashing, see the ThirdReality USB Dongle Guide